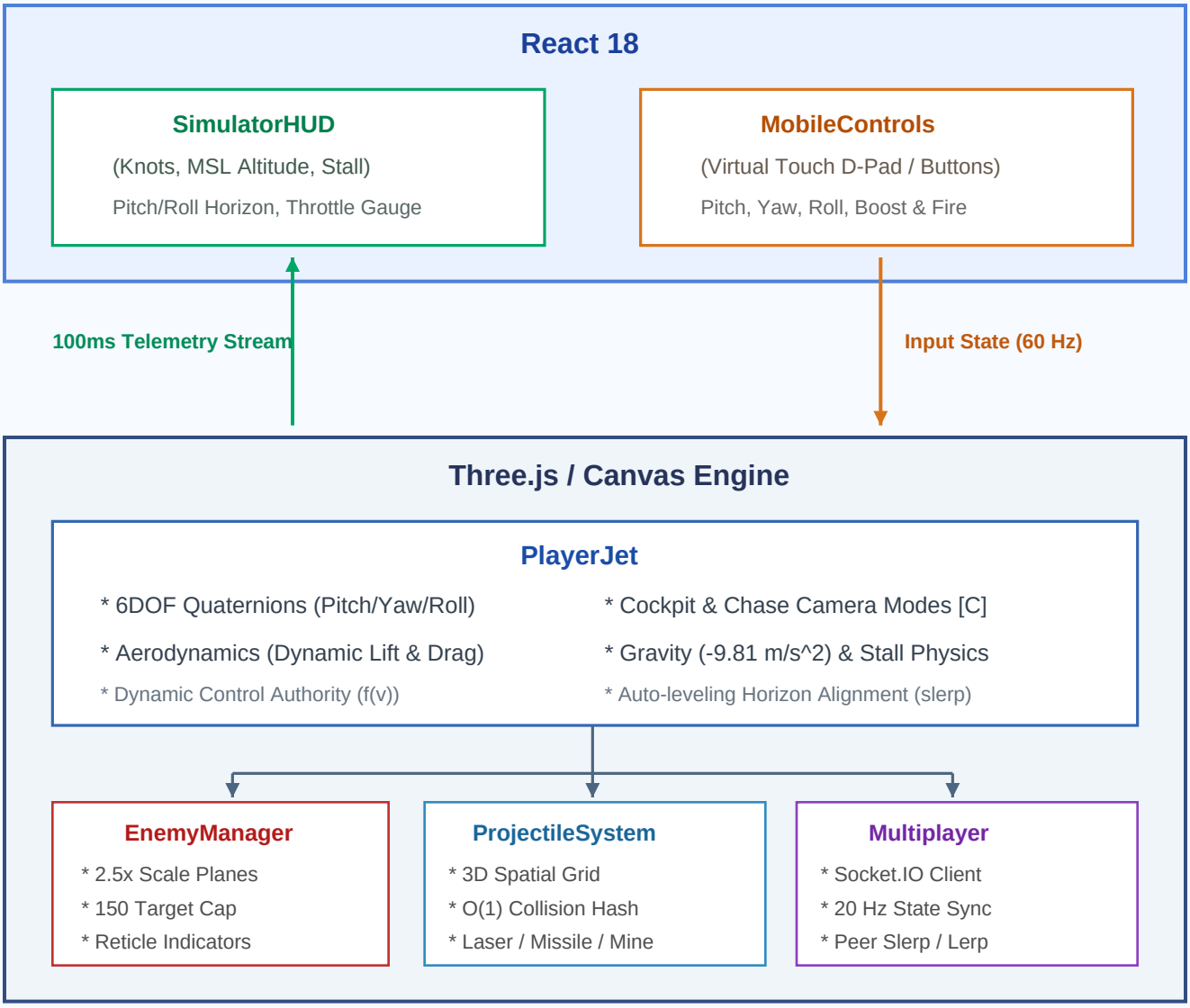
JET STREAK - FIGHTER JET SIMULATOR

Runtime Architecture & Component Context

SYSTEM RUNTIME ARCHITECTURE



VERBATIM CONTEXT SPECIFICATION

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+-----------------------------------------------------+
|                      React 18                       |
|   +-------------------+      +--------------------+ |
|   |   SimulatorHUD    |      |   MobileControls   | |
|   | (Knots, MSL, Stall)|     |  (Touch D-Pad/Btns)| |
|   +---------^---------+      +---------v----------+ |
+-------------|--------------------------|-------------+
              | 100ms Telemetry          | Input State
+-------------v--------------------------v-------------+
|                Three.js / Canvas                    |
|  +-------------------------------------------------+ |
|  |                   PlayerJet                     | |
|  |  - 6DOF Quaternions     - Cockpit / Chase Cam   | |
|  |  - Aerodynamics (L/D)   - Stall & Gravity Physics| |
|  +----------------------v--------------------------+ |
|                         |                           |
|     +-------------------+--------------------+      |
|     v                   v                    v      |
| EnemyManager      ProjectileSystem     Multiplayer  |
| (2.5x Scale,      (3D Spatial Grid,    (Socket.IO,  |
| 150 Target Max)   O(1) Broadphase)     Peer Lerp)   |
+-----------------------------------------------------+
```

TECHNICAL SUBSYSTEM SPECIFICATIONS

In-Depth Breakdown of Flight Mechanics, Combat & Networking

1. Aerodynamics & 6DOF Flight Physics Model

Implemented in src/systems/PlayerJet.js:

* **Net Force Vector Equation:** F_net = (Thrust - Drag) * Forward + Lift * Up + Gravity
* **Dynamic Lift & Drag:** Lift = v^2 * LIFT_COEFF (0.0028); Drag = v^2 * DRAG_COEFF (0.0008).
* **Stall Dynamics:** Airspeed threshold at 22 m/s (~44 knots). 80% loss of lift on stall, inducing continuous downward pitch recovery torque until pilot dives to regain airspeed.
* **Dynamic Surface Authority:** Control responsiveness scales with sqrt(v / v_cruise), preventing sudden twitchy movements at low airspeeds and stiffening control surfaces during high-speed dives.
* **Dual Camera Perspectives:** Cockpit (first-person internal pilot view) and Chase (trailing 3rd person). Switchable instantaneously via the C key or HUD camera toggle.

2. UI Avionics & Responsive Touch Controls

* **Telemetry Pipeline:** GameCanvas exposes calibrated getters to App.jsx on a 100ms ticker interval: getAirspeedKnots() [speed * 2.2], getAltitudeFeet() [(pos.y + 50) * 3.28], getThrottlePercent().
* **SimulatorHUD Overlay:** Renders altitude/speed vertical tape gauges, horizon boresight, and stall alert banner.
* **Mobile Touch System:** Virtual on-screen D-Pad handling 8-way directional vector pitch/yaw/roll and dedicated action triggers for boost thrust and cannon fire.
* **Zero DOM Drag Overhead:** Decoupled DOM overlays render outside WebGL Canvas with pointer-events: none to eliminate interference with Three.js rendering performance.

3. Dogfight Scaling, Spatial Grid & Real-time Multiplayer

* **3D Hash Spatial Partitioning:** SpatialGrid partitions 3D space into 50x50x50 unit buckets. Bullets query only nearby buckets, dropping broadphase collision cost from O(N * M) to O(1).
* **Intensive Dogfight Capacity:** Enemy pool scaled to 150 active fighters with 2.5x enlarged mesh size and overhead reticle rings for sharp engagement contrast at supersonic speeds.
* **Multiplayer Synchronization:** Socket.IO backend broadcasts positional coordinates, quaternions, speed, and weapon discharges at 20 updates per second (50ms interval).
* **Peer Smoothing (Lerp / Slerp):** Remote aircraft interpolate positions via lerp and orientations via quaternion slerp, delivering fluid 60 FPS motion without network stutter.
* **Procedural Audio Synthesis:** Zero sound asset files required. AudioSystem synthesizes dynamic engine hums, missile whooshes, and noise-buffer explosions directly via the Web Audio API.